

JACOB PRINZ

Website: jeprinz.github.io

Email: jprinz@umd.edu

EDUCATION

University of Maryland, College Park

August 2020 - (planned) 2026

PhD in Computer Science

Title: Typed Embeddings for Metaprogramming

Advisor: Leonidas Lampropoulos

University of Maryland, College Park

August 2016 - May 2020

B.S. in Mathematics

WORK EXPERIENCE

Indiana University Mathematics REU Program

Summer 2018

Advisor: Noah Snyder

Studied “Homotopy Type Theory,” a field at the intersection of mathematics and computer science. Calculated homotopy groups of spheres in Homotopy Type Theory.

University of Maryland, Losert Lab

Fall 2018

Advisor: Anton Peshkov

Wrote image processing code for a granular physics experiment

Fermilab and University of Maryland

Summer 2017

Advisor: Sarah Eno

Wrote data analysis code for a particle physics experiment. Discovered a bug in a particle physics library.

Catt Labs

Summer 2014 and 2015

Software Developer

Worked as a software developer for two summers. Wrote Java, Javascript, and PHP to develop a server and web applications to disseminate traffic data.

PUBLICATIONS

- A Monad for Nonconstructivity** Preprint
Jacob Prinz and Leonidas Lampropoulos
- Pantograph: A Fluid and Typed Structure Editor** POPL 2025
Jacob Prinz, Henry Blanchette, and Leonidas Lampropoulos
- Merging Inductive Relations** PLDI 2023
Jacob Prinz and Leonidas Lampropoulos
- Deeper Shallow Embeddings** ITP 2022
Jacob Prinz and Leonidas Lampropoulos

BOOKS

- Mathematical Muffin Morsels: Nobody Wants a Small Piece** May 2020
William Gasarch, Erik Metz, **Jacob Prinz**, and Daniel Smolyak

TALKS

- Merging Inductive Relations NJPLS May 2023
Jacob Prinz
- Invited Talk on Deeper Shallow Embeddings Portland PLV Seminar Apr 2023
Jacob Prinz
- Zypr: A New Approach to Structural Editing NJPLS Oct 2022
Henry Blanchette, Jacob Prinz
- Deeper Shallow Embeddings NJPLS May 2022
Jacob Prinz

TEACHING

- UMD CMSC631 **Program Analysis and Understanding** Spring 2025
Teaching Assistant
Professor: Leonidas Lampropoulos
- UMD CMSC433 **Programming Language Technologies and Paradigms** Fall 2021
Teaching Assistant
Professor: Anwar Mamat
- UMD CMSC456 **Introduction to Cryptography** Fall 2018
Teaching Assistant
Professor: William Gasarch
- UMD MATH299T **Homotopy Type Theory** Spring 2019
Student Initiated Course Lecturer
Student Initiated Course Supervisor: Sarah Yeakel